CDS 6324

DATA VISUALIZATION

PROJECT (40%)

Lecture Section: TC2L

Tutorial Section: TT2L

Group Number: TT2L_G12

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1. Data Sources

Dataset Name: Global STI Intelligence: HIV · Gonorrhea · Syphilis

Dataset Source: https://www.kaggle.com/datasets/kanchana1990/global-sti-intelligence-hivgonorrhea-syphilis

The dataset used in this project is the Global STI Intelligence: HIV, Gonorrhea, and Syphilis dataset. It contains global sexually transmitted infection indicators related to HIV, gonorrhea, and syphilis. The original dataset consists of 67,438 records and 31 attributes, covering data from 1990 to 2024 across 200 countries. It includes disease-related information, country and regional information, time-based attributes, numerical health indicator values, burden classifications, confidence interval values, and year-over-year trend indicators.

This dataset is relevant to Sustainable Development Goal 3: Good Health and Well-being because it focuses on global public-health indicators related to sexually transmitted infections. By using country-level and year-based STI data, the dashboard helps users explore disease burden, regional differences, and changes in STI indicators over time. The dataset supports the project goal by allowing users to identify high-burden countries, observe long-term trends, compare diseases, and detect countries with worsening or improving year-over-year values.

The processed dataset is stored as sti_dashboard_clean.csv. It keeps the main dashboard fields, including disease, disease_type, metric, country_code, country_alpha2, country_name, who_region, year, value, confidence interval fields, burden_tier, value_prev_year, yoy_change_abs, yoy_change_pct, and yoy_direction. Two supporting files were also generated: metric_catalog.json and data_dictionary.json. The metric catalog supports the dashboard filter options by listing available metrics by disease and disease type, while the data dictionary explains each cleaned and derived field used in the dashboard.

During preprocessing, numeric fields such as value, value_low, value_high, ci_width, value_prev_year, yoy_change_abs, and yoy_change_pct were cleaned and converted into usable numerical values. Text fields such as disease, disease_type, metric, country codes, country name, WHO region, burden tier, and year-over-year direction were also standardized. Rows outside the 1990 to 2024 range, rows without required country or disease information, and rows without a usable main value were removed.

A key preprocessing decision was the creation of a normalized_score field. The dataset contains many different metrics, and not all metrics use the same unit or meaning. For example, one metric may represent a count, while another may represent a percentage or rate. Directly comparing these raw values across diseases could produce misleading visual conclusions. To avoid this issue, normalized_score was calculated within each disease and metric group using the following formula:

- normalized_score = value / maximum value within the same disease + metric group

The result is capped between 0 and 1. A value closer to 1 means that the record is closer to the highest observed value within the same disease and metric group. Raw metric values are still used when the user selects a specific compatible metric. However, normalized_score is used when the dashboard needs to compare across diseases, such as in Disease = All mode, the Dominant Disease KPI, and the Disease Share Donut Chart.

Another derived field, burden_tier_score, was created to convert burden categories into numeric values. The conversion uses LOW = 1, MEDIUM = 2, HIGH = 3, VERY HIGH = 4, and UNKNOWN as blank or null. This field is useful when a visualization needs a numeric representation of burden severity, such as the animated bubble chart when year-over-year change values are unavailable.

Overall, the dataset is suitable for this project because it satisfies the required dataset conditions, contains both time and location dimensions, and supports the construction of six meaningful interactive D3.js visualizations. The preprocessing stage ensures that the final dashboard uses a cleaner, more consistent, and more responsible version of the original dataset.

Column Name	Description
record_id	Unique identifier for each record
disease	Name of the STI, such as HIV, Gonorrhea, or Syphilis
disease_type	Disease category, such as Viral or Bacterial
disease_treatment	Main treatment approach, such as ART, antibiotics, or penicillin
is_curable	Indicates whether the disease is curable; 1 = curable, 0 = not curable
indicator_code	Code representing the health indicator
metric	Specific public-health measurement or indicator
country_code	Three-letter country code
country_alpha2	Two-letter country code
country_name	Name of the country
who_region	WHO region, such as Africa, Americas, or Europe
year	Year of the observation
decade	Decade grouping, such as 1990s or 2000s
is_recent	Indicates whether the record is considered recent
years_ago	Number of years before the dataset scrape date
sex	Original sex or subgroup code used by the source
sex_label	Display label for sex group; in this dataset it is mainly Both Sexes
value	Main numerical value for the selected metric
value_low	Lower estimate or confidence interval value
value_high	Upper estimate or confidence interval value
value_display	Formatted version of the value for display
ci_width	Width of confidence interval
burden_tier	Burden classification such as LOW, MEDIUM, HIGH, or VERY HIGH
value_prev_year	Value from the previous year
yoy_change_abs	Absolute year-over-year change
yoy_change_pct	Percentage year-over-year change
yoy_direction	Direction of change, such as INCREASED, DECREASED, or STABLE
data_source	Original data source name
source_url	URL of the source API or dataset
scrape_date	Date when the dataset was collected
dataset_version	Version of the dataset

2. Analytical Questions

The dashboard was designed around four analytical questions. These questions guide the choice of visualizations and ensure that the dashboard supports meaningful exploration rather than only displaying raw data. Each question is connected to one or more visualizations so that users can move from a general overview to more detailed analysis.

2.1 Which countries and WHO regions have the highest STI burden for HIV, gonorrhea, and syphilis in a selected year?

This question focuses on identifying geographical differences in STI burden. It is mainly answered using the Choropleth World Map, Ranked Country Bar Chart, and KPI cards. The choropleth map allows users to observe global spatial patterns and identify countries with higher selected metric values or normalized burden scores. The ranked country bar chart complements the map by showing the top countries in a clearer ranked format, making country-to-country comparison more precise. The KPI cards provide a quick summary of the highest and lowest burden countries based on the selected filters.

This question is important because spatial comparison helps users understand where STI indicators are more concentrated. By filtering by metric, disease, year, WHO region, and burden tier, users can examine whether high-burden areas are concentrated in certain regions or distributed globally.

2.2 How have STI indicators changed from 1990 to 2024 across countries and diseases?

This question focuses on long-term temporal trends. It is answered using the Multi-Line Time-Series Chart and the Animated Bubble Chart. The time-series chart shows how selected STI indicators change over the 1990 to 2024 period. Depending on the selected filters, the chart can compare diseases, countries, or WHO regions over time. This helps users observe whether selected indicators are increasing, decreasing, or fluctuating.

The animated bubble chart supports the same question from a more dynamic perspective. It allows users to play through years, pause at a specific year, or manually control the year slider. This makes temporal progression easier to observe, especially when comparing multiple countries or regions. Together, the time-series and animated bubble chart help users understand how STI burden changes over time rather than only showing a single-year snapshot.

2.3 Which countries show the fastest worsening or improvement in STI indicators based on year-over-year change?

This question focuses on change detection. It is answered using the Year-over-Year Lollipop Chart and the Fastest Worsening Country and Fastest Improving Country KPI cards. The lollipop chart uses year-over-year percentage change to highlight countries with the largest increases and decreases for the selected metric and year.

Positive year-over-year change values indicate that the selected indicator has increased, while negative values indicate that it has decreased. The vertical zero line in the lollipop chart separates worsening and improving directions, making the comparison easier to understand. The KPI cards summarize the most extreme changes so that users can immediately identify the country with the strongest increase and the country with the strongest decrease under the current filter settings.

This question is useful because countries with rapid changes may require further investigation. A country with high burden may already be visible on the map, but a country with a sudden increase or decrease may only become clear through year-over-year analysis.

2.4 Which STI contributes the largest share of the selected health indicator under different filters?

This question focuses on disease composition. It is answered using the Disease Share Donut Chart and linked dashboard filters. The donut chart shows the relative contribution of HIV, gonorrhea, and syphilis under the current filter context.

Because the dataset contains different metrics with different units and meanings, the donut chart uses normalized burden score instead of directly comparing raw values across diseases. This avoids misleading comparisons between incompatible measurements. The donut chart is therefore labelled as "Disease Share by Normalized Burden Score." Users can click a disease segment to filter the dashboard and examine that disease in more detail across the other visualizations.

This question helps users understand which disease has the largest relative contribution under selected conditions, such as a specific region, year, burden tier, or metric context. It also connects disease-level analysis with the spatial, ranking, temporal, and year-over-year views.

Overall, these four analytical questions form the dashboard story. The first question identifies where STI burden is highest, the second explains how the indicators change over time, the third highlights where the fastest changes occur, and the fourth shows how disease contribution differs under selected filters. Together, they support a structured exploration of global STI burden in relation to SDG 3: Good Health and Well-being.

3. Visualizations

3.1 KPI Cards

The KPI cards provide a quick summary of the current dashboard context before users inspect the detailed charts. The six KPI cards are Global Burden Score, Highest Burden Country, Lowest Burden Country, Dominant Disease, Fastest Worsening Country, and Fastest Improving Country.

Their purpose is to highlight the most important values based on the selected filters. For example, when users change the disease, metric, year, country, or WHO region, the KPI cards update automatically. This helps users quickly identify the highest and lowest burden countries, the dominant disease by normalized burden score, and countries with the strongest year-over-year increase or decrease.

Key insight supported: The KPI cards help users immediately understand the main summary points of the selected data context without needing to inspect every chart individually.

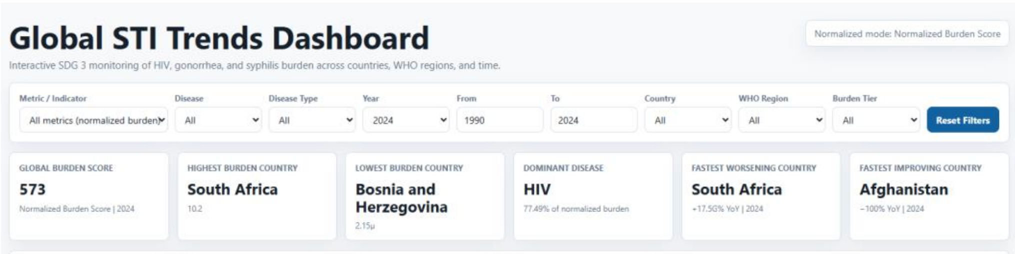

Figure 3.1: Dashboard filters and KPI cards

3.2 Choropleth World Map

The Choropleth World Map shows STI burden by country. Its purpose is to reveal geographical patterns and identify countries or regions with higher burden under the selected metric, disease, year, and filter settings.

The main visual encoding is color intensity. Countries with higher selected values or normalized burden scores appear with stronger color intensity, while countries without matching data appear in neutral grey. This allows users to quickly compare spatial patterns across the world.

The map supports hover tooltips, zooming, panning, reset zoom, and country selection. When a country is clicked, the other dashboard charts update to focus on that country. This makes the map useful as both a spatial overview and an entry point for detailed country-level analysis.

Key insight supported: Users can identify where STI burden is geographically concentrated and then select a country for deeper analysis.

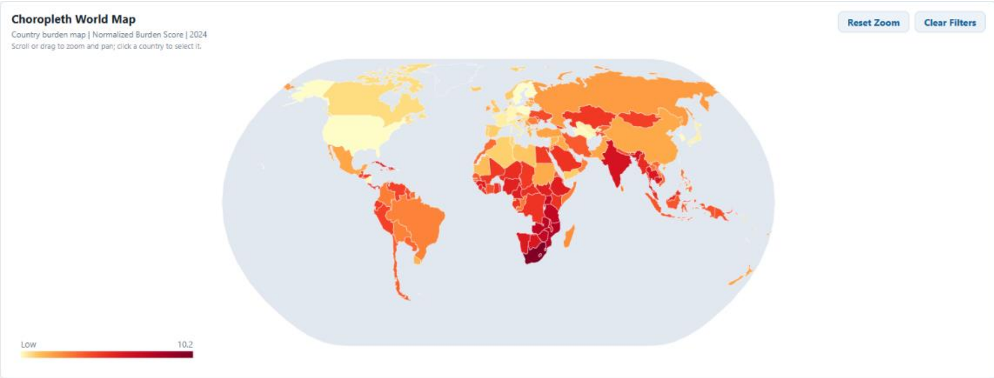

Figure 3.2: Choropleth world map showing country-level STI burden.

3.3 Ranked Country Bar Chart

The Ranked Country Bar Chart compares countries using a top-ranking format. Its purpose is to show which countries have the highest selected metric values or normalized burden scores in a clearer and more precise way than the map.

The main visual encoding is bar length. Longer bars represent higher values, and the countries are sorted in descending order. This makes it easy to compare the top countries and identify the highest-ranked countries under the current filters.

The chart includes hover tooltips showing rank, country, value, disease, year, and burden tier. Clicking a bar selects that country globally and updates the other charts. This supports linked exploration from ranking to country-specific trends and changes.

Key insight supported: Users can clearly compare the countries with the highest STI burden and see exact ranking differences.

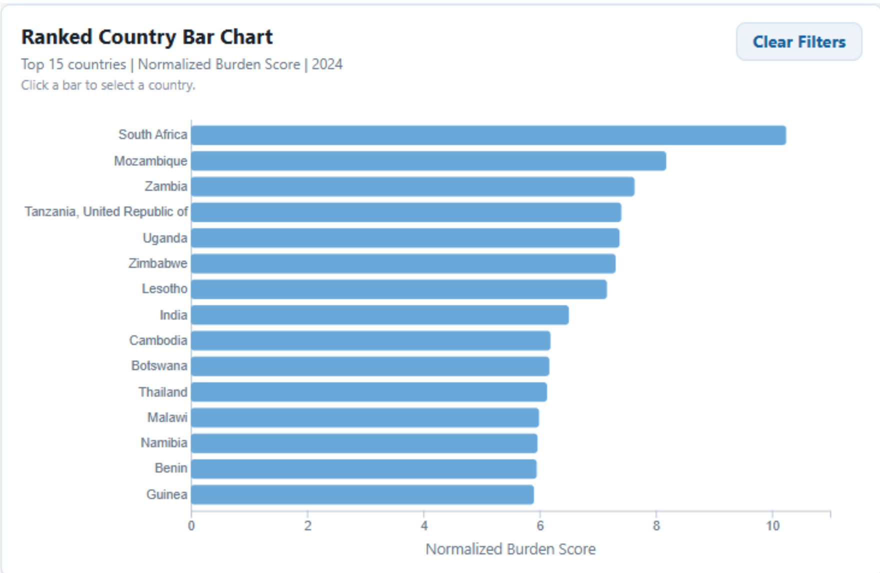

Figure 3.3: Ranked country bar chart comparing top countries

3.4 Multi-Line Time-Series Chart

The Multi-Line Time-Series Chart shows how STI indicators change over time from 1990 to 2024. Its purpose is to support long-term trend analysis across diseases, countries, or WHO regions depending on the selected filter context.

The x-axis represents year, and the y-axis represents the selected raw metric value or normalized score. Lines are grouped by disease or WHO region depending on the current dashboard state. This allows users to observe whether selected indicators increase, decrease, or fluctuate over time.

The chart includes hover tooltips and a brushing feature. Users can brush over a specific time period to focus on a shorter year range and use the reset range control to return to the full 1990–2024 view.

Key insight supported: Users can identify long-term changes in STI indicators and compare how trends differ between selected diseases, regions, or countries.

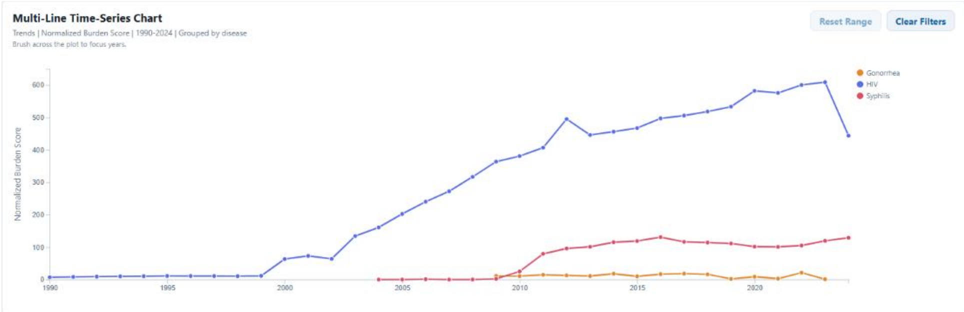

Figure 3.4: Multi-line time-series chart showing STI indicator trends

3.5 Animated Bubble Chart

The Animated Bubble Chart shows country-level STI burden changes across years. Its purpose is to provide a dynamic view of temporal progression and satisfy the project requirement for meaningful animation.

Bubble size represents the selected value mode, such as raw metric value or normalized burden score. Bubble color represents disease or WHO region depending on the filter context. The chart includes Play, Pause, and year-slider controls, allowing users to move through the years automatically or manually.

Hover tooltips display country, disease, WHO region, year, value, year-over-year change, and burden tier. Clicking a bubble selects a country and updates the other charts. The animation is analytical because it helps users observe how selected countries or groups change over time.

Key insight supported: Users can observe year-by-year changes in country-level STI burden and compare how countries move across different years.

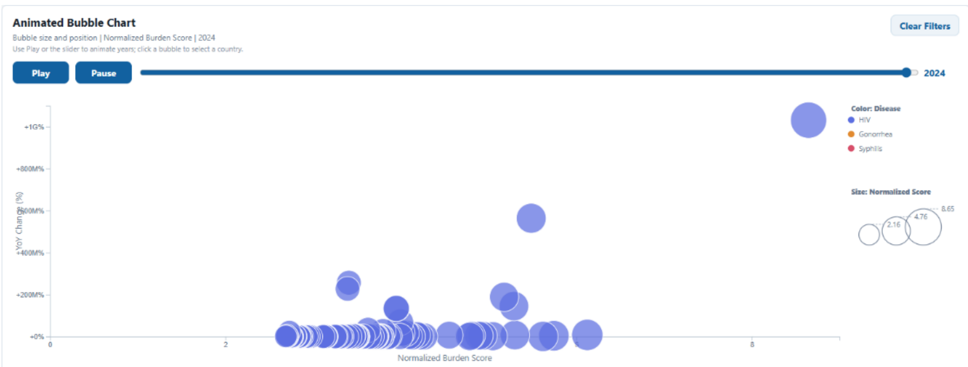

Figure 3.5: Animated bubble chart showing temporal country-level changes

3.6 Year-over-Year Lollipop Chart

The Year-over-Year Lollipop Chart highlights countries with the strongest increase or decrease based on year-over-year percentage change. Its purpose is to identify countries with the fastest worsening or improving STI indicators.

The chart uses yoy_change_pct as the main measure. Positive values indicate increases, while negative values indicate decreases. A zero line separates worsening and improving directions, making the direction of change easy to understand.

Hover tooltips show previous value, current value, year-over-year percentage change, and direction. Clicking a dot selects the country globally. This chart is useful because rapid changes may not always be visible from absolute burden values alone.

Key insight supported: Users can identify countries with the fastest year-over-year worsening or improvement for the selected STI indicator.

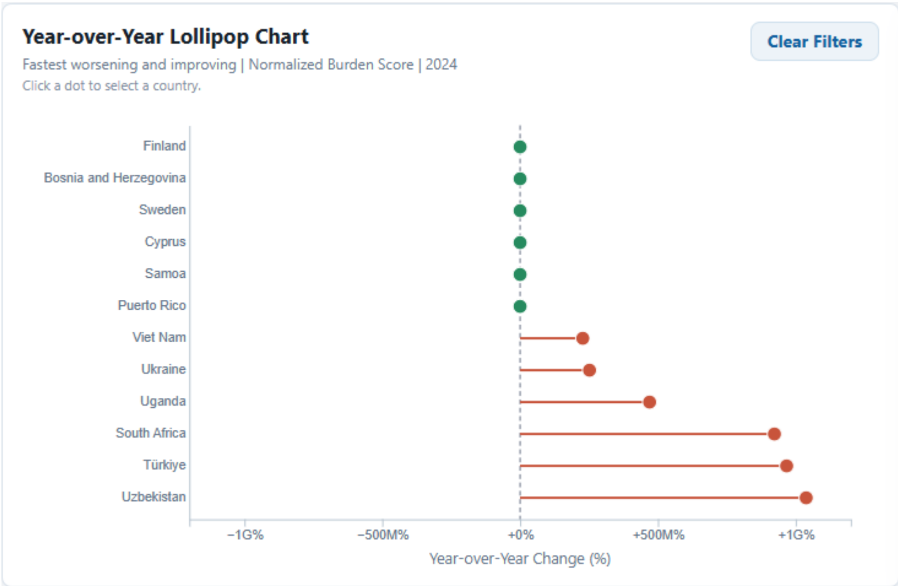

Figure 3.6: Year-over-year lollipop chart showing fastest worsening and improving countries

3.7 Disease Share Donut Chart

The Disease Share Donut Chart shows the relative contribution of HIV, gonorrhea, and syphilis under the selected filters. Its purpose is to summarize disease composition and identify which disease contributes the largest relative share. The chart uses normalized burden score rather than raw values. This is important because the dataset contains many different metrics with different units and meanings. Using normalized score avoids misleading comparisons across diseases. Each donut segment represents one disease, and segment size represents its share of the total normalized burden score. Hover tooltips show the disease, normalized burden share, and percentage. Clicking a segment filters the dashboard by disease.

Key insight supported: Users can understand which STI contributes the largest relative share under the selected data context while avoiding unfair raw-value comparisons.

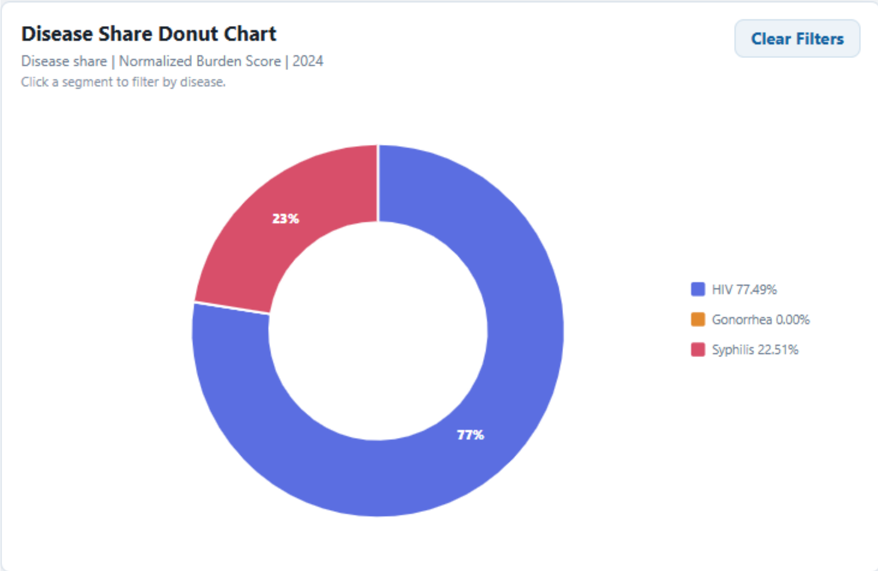

Figure 3.7: Disease share donut chart using normalized burden score

4. Interactivity

The dashboard uses interactive features to help users explore the STI dataset more effectively. These interactions allow users to filter the data, inspect details, focus on specific countries or years, and observe changes over time.

The main interaction is the global filter panel. Users can filter the dashboard by Metric / Indicator, Disease, Disease Type, Year, Year Range, Country, WHO Region, and Burden Tier. When a filter changes, all KPI cards and charts update automatically. A Reset Filters button restores the dashboard to its default state.

All six visualizations include hover tooltips. The tooltips display useful details such as country, WHO region, disease, year, selected metric, value, burden tier, rank, year-over-year change, or normalized burden share. This keeps the charts clean while still allowing users to inspect exact information.

The Choropleth World Map supports zooming and panning, allowing users to examine smaller countries and dense regions more clearly. It also includes a Reset Zoom button. The Multi-Line Time-Series Chart includes brushing, which lets users select a shorter year range, with a Reset Range button to return to 1990–2024.

Linked interactions connect the dashboard views together. Clicking a country on the map, ranked bar chart, animated bubble chart, or lollipop chart selects that country and updates the other charts. Clicking a disease segment in the donut chart filters the dashboard by that disease. These interactions help users move from overview to detailed analysis without manually changing every filter.

The Animated Bubble Chart includes Play, Pause, and year-slider controls. This animation is meaningful because it shows how country-level STI indicators change across years. Users can either watch the yearly progression automatically or select a specific year manually.

Each chart also includes a Clear Filters button, making it easier to reset the dashboard during testing or presentation. Empty-state messages are displayed when a selected filter combination has no available data. Overall, the interactive features support flexible exploration and help answer the analytical questions through filtering, inspection, selection, brushing, zooming, and animation.

5. Best Data Visualization Practices

The dashboard design was guided by Shaffer's 4Cs: Clear, Clean, Concise, and Captivating. These principles were used to ensure that the visualizations communicate meaningful insights without overwhelming users.

The first principle is clarity. Each chart has a specific purpose and is matched to the type of analysis it supports. The choropleth map shows geographical distribution, the ranked bar chart compares countries, the time-series chart shows long-term trends, the animated bubble chart shows yearly progression, the lollipop chart highlights year-over-year change, and the donut chart shows disease share. Chart titles, axis labels, legends, and tooltips are included to help users understand what is being shown.

The second principle is cleanliness. The dashboard avoids unnecessary decoration and focuses on the data. Rankings are limited to the top countries to reduce clutter, and detailed values are placed in hover tooltips instead of being shown permanently on the charts. Countries with no map data are shown in neutral grey so they do not distract from valid data.

The third principle is conciseness. The dashboard uses KPI cards to summarize the most important values quickly, while the six main charts provide more detailed exploration. This allows users to understand the overall situation first and then investigate specific countries, diseases, years, or regions.

The fourth principle is consistency. Similar colors, labels, tooltip formats, and interaction patterns are used across the dashboard. For example, disease and region categories are represented consistently, and clicking a country or disease updates the dashboard in a predictable way. This helps users learn the interface quickly.

The dashboard also follows responsible comparison practices. Since the dataset contains many different metrics with different units and meanings, raw values are not silently compared across diseases. Raw values are used when a specific compatible metric is selected, while normalized_score is used for cross-disease views such as the Disease Share Donut Chart and Dominant Disease KPI. This improves graphic integrity and avoids misleading interpretation.

Overall, the dashboard applies clear chart selection, consistent visual encoding, readable labels, useful tooltips, and careful comparison rules to support accurate and effective data exploration.

6. Challenges & Solutions

Several challenges were faced during the development of the dashboard. The first challenge was the size of the raw dataset. The original CSV file was large and contained many fields that were not needed for visualization. Loading the full raw file directly in the browser could reduce dashboard performance. To solve this, a Python preprocessing script was used to generate a smaller dashboard-ready dataset containing only the required fields.

The second challenge was the presence of many different metrics in the dataset. Some metrics represent counts, while others represent rates or percentages. Comparing these raw values directly across diseases could be misleading. To address this, the dashboard includes a Metric / Indicator filter, and cross-disease views use normalized_score instead of raw values. This helps preserve graphic integrity when comparing HIV, gonorrhea, and syphilis.

Another challenge was joining country data to the world map. The map uses country geometry, while the dataset uses country codes and country names. ISO-3 country codes were used where possible to improve matching accuracy. Countries without matching data are shown in neutral grey rather than being forced into incorrect matches.

The year-over-year change values also created challenges because some percentages can be very large when the previous year's value is close to zero. Instead of removing these records, the dashboard keeps them as part of the dataset but formats the values compactly and provides details through tooltips.

A further challenge was visual clutter, especially in ranking, time-series, and legend-heavy charts. This was addressed by limiting ranking charts to top countries, using external legends where suitable, and moving detailed values into hover tooltips. This keeps the dashboard readable while still allowing users to inspect exact information.

Finally, coordinating interactions across six D3 visualizations was challenging. To solve this, the dashboard uses a shared state structure. Filters and chart selections update the same state, allowing all charts and KPI cards to refresh consistently when users select a country, filter a disease, change a year, or reset the dashboard.

7. Development Process and Group Contribution

The project was developed in stages. First, the team reviewed the project requirements and confirmed that the dashboard would follow the submitted proposal. The Global STI Intelligence dataset was selected because it supports the SDG 3 theme and contains country, region, disease, metric, and year-based attributes. Next, the dataset was cleaned and processed into a dashboard-ready file to improve performance and avoid loading unnecessary raw data into the browser.

After preprocessing, the team developed the dashboard layout, global filters, KPI cards, and individual D3.js visualizations. Each member was responsible for two main visualizations, while shared work included integration, testing, report writing, and presentation preparation. The visualizations were then connected through shared dashboard state so that filters and chart selections could update the whole dashboard consistently.

Collaboration was managed by dividing the work according to the proposal. Yaser focused on temporal analysis through the Multi-Line Time-Series Chart and Animated Bubble Chart, as well as dashboard integration. Chan focused on spatial and ranking analysis through the Choropleth World Map and Ranked Country Bar Chart, as well as global filters. Mishal focused on year-over-year change and disease composition through the Year-over-Year Lollipop Chart and Disease Share Donut Chart, as well as linked interactions.

The main challenges during development were cleaning the dataset, handling different metric types, reducing visual clutter, and ensuring that all charts updated correctly when filters changed. These issues were solved through preprocessing, normalized_score calculation, clear chart design, and shared dashboard state. The final stage involved testing the dashboard, preparing screenshots, completing the report, and planning the final presentation.

Table 7.1: Group contribution by members

Name	Student ID	Contribution
Yaser E H Abulaban	1221305612	• Temporal STI trend analysis • Multi-Line Time-Series Chart • Animated Bubble Chart • Dashboard integration • Report writing and presentation preparation
Chan Ga Wai	1221305898	• Spatial and country ranking analysis • Choropleth World Map • Ranked Country Bar Chart • Global filters • Report writing and presentation preparation
Mishal Mann Nair	1221305145	• Year-over-year change and disease composition analysis • Year-over-Year Lollipop Chart • Disease Share Donut Chart • Linked interactions • Report writing and presentation preparation